Conditional Verification for Reliable Medical-Image Classification

Exploratory research prototype | OrganAMNIST | 12 November 2026

Research question

Can a lightweight decision policy selectively invoke a second image classifier and request review, reducing verification effort while retaining high accuracy among accepted predictions?

Experimental design

The prototype uses the OrganAMNIST 11-class organ classification benchmark. An original CNN makes the first prediction; a separately trained verifier CNN can be invoked conditionally. The policy accepts the original prediction without verification unless its softmax confidence is below 0.65 or its predicted class is 6 or 10. When verification is invoked, agreement leads to acceptance and disagreement to review. The reported comparison uses 17,778 test images. Verification-call rate is a proxy for workload, not a measured runtime or cost.

Reported test results

Strategy	Coverage	Accepted accuracy	Verifier calls
Original only	100.00%	80.33%	0.00%
Always verify	83.13%	93.48%	100.00%
Conditional verification	84.67%	92.22%	52.24%

Conditional verification retains 84.67% coverage at 92.22% accuracy among accepted cases, compared with 83.13% coverage and 93.48% accepted accuracy for always verifying. These are selective-prediction metrics: they must not be interpreted as accuracy across all test images.

Reliability and failure analysis

The original model achieved 90.51% accuracy on validation and 80.33% on test, indicating a substantial validation–test performance gap. Of 336 high-confidence (≥0.90) original-model errors on test, disagreement detected 130 (38.69%) and missed 206 (61.31%). Thus, a second model can identify some errors but agreement cannot establish correctness. Previous error-correlation analysis also found substantial shared failure modes between the two CNNs.

Scientific interpretation

The prototype illustrates a transparent tool-use and abstention mechanism: a logged policy decides whether to invoke another model and whether to accept or flag an image for review. Its main value is the measurable trade-off between verification frequency, accepted-case accuracy, and coverage, together with evidence of high-confidence shared errors. It is a rule-based conditional verification workflow, not a clinically deployed autonomous agent.

Limitations and validity

• Policy development was informed by prior test-set analysis; test results are exploratory and not an independent final estimate.

• The task is 28×28 organ classification, not tumour segmentation, longitudinal imaging, or treatment-response assessment.

• Softmax confidence is not necessarily calibrated; model agreement is not a correctness guarantee.

• The two models may share systematic errors; a more diverse verifier and independent external test data are needed.

• No clinical validation, real-world review study, measured latency, or patient-level outcomes are reported.

Next research steps

Pre-register the decision rule on a development split and evaluate it once on a genuinely untouched holdout or external dataset; measure risk–coverage curves, class-wise error, calibration, and runtime; compare against confidence-only and disagreement-only baselines; test shift robustness and verifier diversity. A later research extension could replace organ classification with longitudinal tumour-imaging measurements and auditable, uncertainty-aware tool decisions.

Source and reproducibility note

Prepared from the supplied experiment_report.json, conditional_benchmark.csv, and corrected compare_validation_models.py, supplemented by earlier terminal results. Reported metrics have not been independently reproduced from model checkpoints or raw predictions. The experiment JSON records exploratory status and prior test-informed policy development.